PROJECT PROPOSAL

AUTOMATED THREAT INTELLIGENCE & TRIAGE ORCHESTRATOR

(Project: AI-AUGMENTED INCIDENT REPORTING ENGINE (AIRE) FOR WAZUH)

Submitted To:
The Faculty Committee
Cyber Academy Mysore

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1. Executive Summary

In the rapidly evolving landscape of cyber threats, speed is the ultimate currency. This proposal introduces the AI-Augmented Incident Reporting Engine (AIRE), a sophisticated orchestration framework designed to bridge the gap between threat detection and actionable response within the Wazuh ecosystem. By synergizing Large Language Models (LLMs) with automated threat intelligence, AIRE transforms the traditional, labor-intensive reporting process into a near-instantaneous, high-fidelity narrative. Our mission is to move beyond simple alerting, providing security teams with a tool that thinks, enriches, and reports with the precision of a veteran analyst.

2. Strategic Problem Statement

Modern Security Operations Centers (SOCs) are currently hindered by "Analysis Paralysis." While Wazuh excels at identifying potential threats, the subsequent human workflow remains a bottleneck. Security analysts are often submerged in:

- **Data Fragmentation:** Manually correlating logs across disparate endpoints.
- **Contextual Voids:** A lack of immediate external intelligence (IP reputation, malware hashes).
- **Documentation Fatigue:** Spending more time drafting reports for compliance than actually neutralizing threats.

Currently, documenting a single critical breach can consume up to an hour of an analyst's time. In a high-stakes environment, this delay is a luxury we cannot afford. AIRE is engineered to reclaim this time.

3. The AIRE Vision: Operational Architecture

AIRE is not a replacement for the human analyst; it is a force multiplier. It integrates seamlessly into the Wazuh pipeline to provide a modular, event-driven response for Level 12+ (Critical) alerts.

The Workflow Logic:

1. **Intelligent Triggering:** Utilizes Wazuh's integratord to isolate high-impact events.
2. **Contextual Enrichment:** Automatically queries global threat databases to validate the severity of the alert.
3. **Neural Synthesis:** Employs LLMs to digest complex JSON data and generate a narrative that identifies the "Who, What, and How."
4. **Actionable Delivery**: Pushes finished reports directly into the hands of decision-makers via Slack, Teams, or Jira.

4. Technical Specifications

To ensure enterprise-grade reliability, AIRE utilizes a modern, robust tech stack:

- **Platform:** Wazuh Manager 4.7+ (Open-Source Core).
- **Language:** Python 3.10+ (utilizing FastAPI for high-concurrency handling).
- **Intelligence APIs:** VirusTotal, AbuseIPDB (Community and Premium tiers).
- **AI Engine:**
    - ***Cloud:*** OpenAI GPT-4o-mini (for high-speed reasoning).
    - ***On-Prem:*** Ollama/Llama 3 (for environments requiring absolute data privacy).
- **Knowledge Base**: FAISS or ChromaDB for RAG-based policy alignment.

5. Technical Approach & Innovation

AIRE introduces three core innovations to the standard Wazuh deployment:

- **Retrieval-Augmented Generation (RAG): By** embedding your organization's internal Standard Operating Procedures (SOPs) into a vector database, AIRE ensures that every report generated is strictly aligned with local security policies.
- **Autonomous Research Agents:** Using LangChain, the system acts as a "Digital Researcher," investigating suspicious hashes and IPs before the analyst even opens the alert.
- **Alert Throttling:** Smart buffering prevents "Alert Storms," grouping related events into a single comprehensive incident story.

6. Implementation Roadmap: A 10-Week Accelerated Development Cycle

This project is structured into five distinct bi-weekly sprints, ensuring a transition from a raw data stream to a sophisticated AI-driven intelligence system within one academic quarter.

Phase	Duration	Focus	Key Milestone
I. Foundation	**Weeks 1–2**	**Infrastructure & Interception:** Establishing the bridge between the Wazuh Manager and the Python environment.	Successful real-time capture of Level 12+ alerts via ossec-integratord.
II. Enrichment	**Week 3**	**Contextual Augmentation:** Integrating external Threat Intelligence (OSINT) to validate raw logs.	Real-time API synthesis (VirusTotal/AbuseIPDB) and data cleaning logic.
III. Intelligence	**Weeks 4–5**	**Cognitive Logic & RAG:** Prompt engineering for LLMs and building the Vector Database for Policy Alignment (RAG).	Successful conversion of complex JSON logs into structured human-readable narratives.
IV. Delivery	**Weeks 6–7**	**The Output Ecosystem:** Automating the generation of professional-grade reports and cross-platform delivery.	Automated generation of Markdown/PDF reports and live Slack/Jira integration.
V. Validation	**Weeks 8–10**	**Stress Testing & Hardening:** Simulating "Alert Storms" and optimizing processing speed.	System audit showing <15 second report generation under peak load and final documentation.

7. Impact Assessment: The AIRE Strategic Advantage

The deployment of AIRE shifts the security paradigm from *reactive log digging* to *proactive incident management*.

Metric	Traditional SOC Workflow	AIRE-Enhanced Workflow	Improvement
Initial Report Latency	30 – 60 Minutes (Manual)	**< 15 Seconds (Autonomous)**	**~99% Reduction**
Data Fidelity	Variable based on Analyst experience	**Consistent, Standardized, & Data-Driven**	High Standardization
Contextual Depth	Internal logs only (Manual OSINT)	**Pre-enriched with Global Threat Intel**	Instant Intelligence
Compliance Readiness	Hours of manual mapping to frameworks	**Instant MITRE ATT&CK & GDPR Mapping**	Automated Compliance
Analyst Efficiency	80% Documentation / 20% Response	**5% Review / 95% Remediation**	Focus Shift to Action

8. Conclusion: Redefining the Frontline of Defense

The **AI-Augmented Incident Reporting Engine (AIRE)** is more than a technical integration; it is a strategic evolution in the fight against sophisticated cyber adversaries. By condensing the critical hour of manual data gathering into a few seconds of AI-driven synthesis, AIRE effectively solves the primary challenge of the modern SOC: **The Time-Gap.**

Over a focused **10-week development timeline**, this project will transform Wazuh from a detection platform into an intelligence powerhouse. We are not just automating a report; we are providing security defenders with "Just-in-Time" intelligence—ensuring that when a critical breach occurs, the response is immediate, informed, and insurmountable.

We are prepared to demonstrate that with AIRE, the **Cyber Academy Mysore** and the broader **Cyber Lancers** community can set a new global standard for AI-driven SecOps.